

Schematic completion status

PetSafe 100-1339 R03 A | v0.9.26 | 2026-10-09

All catalog entries are drawn on one sheet. The remaining questions are listed explicitly below. This is a partially verified reconstruction, not a completed hardware validation or a routed PCB.

Coverage	Open connections	Values / packages
149 entries; 1 A2 sheet; 22 stock + 4 custom symbol definitions.	0 populated-entry pads + 2 DNP pads. 0 open PIC pads in the model.	44 ceramic and 2 magnetic values unknown; 148/149 footprints assigned.

Power rails - current names

Name	Known path
BATTERY+	Battery positive -> Q1 single pad/native3 (candidate drain); TP14/R43.2 on the raw side.
VSYS	Q1.L/native2 -> U5/MX512H pin4; L1.1 and L2.1 supply sides.
VDD	TP103, PIC20/C32, U5 pin1/VCC, V064/V063/V075/V082. D3.R via V082 is user-confirmed.
VREF	TP104, PIC4/RA2/V035/C39, R21.2/V071. U3.5 and R22 D/left/model2 at 1 ohm.
GND	Battery negative and ground copper.

L1/L2 share the VSYS supply side. Their opposite ends are not shorted together. H_RF_VDD, H_LDO_IN, H_RX_VDD and H_PIR_VDD describe local branches; these are not additional confirmed independent supplies. Rail voltages have not been measured.

Recommended next checks

- U7-J3 mapping resolved. TP16-TP11/TP17 rejected at 400 kohm each. PIC21-TP16/Q2 confirmed; mistaken PIC21-VREF withdrawn, TP16-VREF 800 kohm. PIR-to-PIC path resolved. C25-RA0 and C26-RA1 local wiring resolved. C6-C5 parallel wiring is photo-supported; capacitor values remain unknown.
- One complete Q3/C10/C11/C12 tuning cell.
- R41-VPP is now confirmed. Finish RF excitation and receiver-output destinations; only S1 contact behavior remains uncertain in the local button circuit.
- After routing: frequency-sensitive values and one powered idle/scan session; footprints follow.

Validation and limits

85 modeled net partitions match KiCad 10.0.5. ERC retains 9 findings: 2 open pins, 1 isolated labels, 5 undriven power checks and 1 regulator-option output conflict. U6A is DNP; its common output with fitted U6 is retained and documented, not suppressed. No artificial NC/power flags hide gaps. Original photo hashes are unchanged. This proves file consistency, not all physical connections.

Isolated modeled nodes: H_D1_FREE. A PIC pin can have a local net and still lack an onward destination.

Completion milestones

- Topology complete:** Populated components and functional block interfaces have supported connections; candidate functions/polarities are compatible and remaining exceptions individually documented.
- Value/package complete:** Required values and geometry recovered; exact manufacturer/ratings separated from equivalent design choices. Routed PCB remains a separate deliverable.

Supply and diode gaps

E01 - D2 / marking 4P

Known: Marking 4P. Diode readings support R -> S -> L series-junction behavior in circuit. Photo R=upper-left, L=upper-right, S=single lower. User confirms R-C20 lower 1 ohm; L-ANT2/R46 upper; S-R46 lower. R46 unpopulated. S-ANT2 OL; S-GND 700 kohm, S-VREF 1.32 Mohm. Outer pads show high resistance to VDD/GND.

Uncertain: All three D2 local destinations are resolved. Exact part, ratings and physical package numbering remain unknown; internal series topology is probable from in-circuit tests. Other RX_A_PLUS branches remain inferred. The purpose/value of the empty R46 option is unknown; no resistor is fitted.

Next: No further D2 rail or local continuity sweep is needed. Seek a manufacturer 4P marking/package match consistent with the series-junction measurements; retain the explicit equivalent-circuit assumption if identity cannot be recovered. Do not fit or bridge R46.

E03 - RF supply / Q8 / C6

Known: Q8 centre B2/native5 is GND via V113/V115. V114 is U2.T2/native2 GND. Outer Q8 pads are joined by the earlier low-resistance result. Candidate Q8 P-source/native2 is on filtered VSYS; P-gate/native3 is driven by U2 output4 through R8, while U2 uses VDD. C6 now parallels C5 on H_RF_VDD/GND, supported by user proposal and visible supply/ground copper in IMG_2431/IMG_2432.

Uncertain: Q8 source-supply path remains independently qualified. C6 wiring is photo-supported; its value/rating remain unknown. F-P=10 ohm supports a supply-related path but does not establish a direct copper join. Exact Q8 maker remains a candidate. Gate-drive headroom is unverified: a hypothetical source=6V and gate-high=4.5V gives VGS=-1.5V, not guaranteed off for the candidate threshold range. These are illustrative, not measured voltages.

Next: Do not repeat C6-to-C5 local continuity: the parallel connection is adopted from photos. Agent reviews Q8 source copper and the earlier F-P=10 ohm result before selecting any hidden-path check. After passive routing, measure VDD plus Q8 source and gate high/low during scanning, relative to PCB GND, to compute VGS. Waveforms require a scope if available; use PCB GND for the scope reference.

Tuning and local endpoints

E04 - Antenna / five tuning cells

Known: All five PIC control inputs are mapped. Seven capacitor-midpoint comparisons gave OL. User now confirms ANT2 directly to D2.L.

Uncertain: The model uses PNPs with emitter/native2 at GND and base driven through a resistor by nonnegative GPIO. This is not ordinary forward-operated PNP switching. Ground association, pad orientation/type or an RF junction-switching role may be wrong/incomplete. Capacitor-midpoint antenna continuations remain unresolved. ANT2-D2.L is now verified; further tuning/antenna connections remain open.

Next: Finish ONE cell, Q3/C10/C11/C12: map Q3 physical paired/single pads, both terminals of all three capacitors on both sides and the antenna continuation. Start from IMG_2433/IMG_2442 and V107. Compare Q3.L to the established GND point, Q3.R to R13.2, Q3.S to C10.1, and V107 to the relevant C10/C11/C12 junction. Confirm candidate pairs, do not infer them as measured. Use diode tests to constrain transistor type; base identification alone need not distinguish emitter/collector. Then use symmetry for the others. Preserve all seven OL exclusions.

E05 - Button circuit

Known: R41=330 ohm from marking 331; user confirms its right/model1 end reaches J1 VPP/MCLR. Its left/model2 end joins R40 lower, C35 lower and S1 lower-right. S1 upper-right reaches C35 upper, previously tied to GND/V018. R40 opposite end remains ICSP_DAT/PIC28.

Uncertain: S1 left-hand common-pair assignments and released/pressed behavior remain inferred. The reconstructed pressed path is VPP through 330 ohm to GND, with ICSP_DAT reaching the switch node through 10k. Operating purpose is not established by the annotation.

Next: Only verify the switch contact behavior if required: released versus pressed continuity of opposite rows. Do not repeat the annotated C35/R40/R41 joins or R41-to-VPP destination. No broad resistor-to-rail sweep.

E06 - Local-only PIC / discrete nodes

Known: Every PIC pad has a modeled local net. PIC2 reaches C25/V031; PIC14 reaches TP11; PIC18 reaches TP17. D3.S-to-R42 measured 1.2 ohm; exact resistor end not named, pad2 selected from photo. R42.1 already reaches TP4/PIC25/RB4. PIC21/RB0 now reaches TP16/Q2.S, separate from VREF; this PIR route is resolved. C26.1 reaches PIC3/RA1 and the existing R28/R29/TP7 net; C25.1 remains PIC2/RA0. Their model2 pads share GND.

Uncertain: Onward roles of those three nodes, D1.R/H_D1_FREE. Local connectivity is not proof that these are intentionally unused. H_RF_CONTROL has only R6.2/R7.1: its excitation source is absent. H_BAT_SENSE has a series path through R4 to PIC24/RB3, so the missing item is functional interpretation/route confidence, not necessarily another MCU endpoint. H_D3_SIGNAL replaces the misleading H_RX_RF_SENSE name; operating role remains unknown.

Next: Use existing photo/via evidence to select specific endpoints for the local-only PIC nodes, D1.R and H_RF_CONTROL. Review R3-R4-C28 together with PIC24: document a compatible sensing mechanism, without adding an extra sampling wire solely from a net-name assumption.

Other routing and identity gaps

E07 - PIR interface / U6 / J3

Known: J3 pin1 supply previously confirmed; pin3-GND 1 ohm; pin2-R38 K/right/model1 1 ohm, opposite end 10 kohm. Pin2-GND 290 kohm. C2NM supports S-812C33AMC; V067 reaches J3.1/U6A, V063 regulated VDD, and U6 ground is resistance-supported. User maps all three U7 pads to corresponding J3 columns, with opposite square-pad numbering. TP16-to-TP11/TP17 each reads 400 kohm; neither is a direct route. New user report identifies PIC21/RB0 as TP16/Q2.S destination; local Q2 upper-right-to-R38 trace also identified. Conflict resolved: TP16-VREF 0.8 Mohm and user explicitly withdraws PIC21-VREF. PIC21/RB0 now joins TP16/Q2.S. v0.9.20: U6.5-U6.2 measured 1 ohm (B28). User annotation also joins physical L2/native5 to R37 upper/model2 and empty U6A single/input (native U106.3 after the v0.9.24 symbol replacement). U6.4/L1 marked NC by user and appears as an isolated surface stub in original photos. Candidate pin5 is internally NC but externally tied to VIN; pin4 remains unused. U6A is shown as an unpopulated three-pin regulator template, with single input pin3, upper-right output pin2 and lower-right ground pin1; old connector pin numbering is superseded.

Uncertain: Remaining U6 output-branch qualifications, actual PIR voltage, R33 branch and Q2 candidate pin functions remain qualified. Connector ordering and PIR output-to-controller destination are resolved.

Next: Do not repeat PIR-to-PIC routing tests. Agent checks remaining U6/R33 visible copper and candidate device orientation; only residual hidden ties or device behavior merit measurements.

E08 - Receiver / bias / Q7

Known: U3.5 and R22.2 are VREF; R22.1 is GND. U3.6 reaches R23 B/left. R23 C/right, C40 lower and R42 upper are one measured junction (1 ohm). The old C40 detector-output link is withdrawn. Prior C18/TP6/VREF high-resistance exclusions remain; Q7 supply-switch topology is still inferred. v0.9.19 user annotated local copper: R47 left -> U3.1/R27 right; R47 right + C42 lower + R25 lower + R48 right share RX_STAGE_LINK. C42 upper -> GND inferred from continuous copper beside U3.4 in IMG_2429/2434. R25-GND and R48-VREF old assumptions withdrawn. R47/C42 remain DNP. No meter reading supplied.

Uncertain: Remaining receiver bias/filter hypotheses, Q7 pin functions and operating transfer function. C23/R25 now continues into R48 through RX_STAGE_LINK; R42.1-TP4/PIC25 retains its independent evidence.

Next: Agent traces visible receiver copper and checks vias first. Reserve the meter for a hidden onward route or a specific conflict with remote-net evidence. Do not repeat the completed R23/C40/R42/C18/TP6 checks.

E09 - Power input / Q1 / rail voltages

Known: BATTERY+ and VSYS are separate. User confirms Q1.L to U5.4 and both L1/L2 supply sides on VSYS. VDD TP is the separate logic rail. R1A matches FMOS3401A marking.

Uncertain: Exact Q1 maker and gate operation, unmeasured rail voltages, and U1 enable/option branches. Reverse-polarity protection is an interpretation, not a powered test.

Next: Cross-check Q1 gate/R44 and U1 enable wiring first. After routing reconciliation, use one powered session on the original hardware/supplies: BATTERY+, VSYS, VDD, VREF, U6 output, Q8 source/gates and both U5 control inputs, idle and during a triggered scan. Receiver power may be switched. No battery/PIC/RF redesign.

E10 - ICSP protection / D6 / J1

Known: D4/D5=5U supports SD05 TVS candidates. D6=G3 has at least two documented Zener candidates. ICSP signal routes are photo-derived. J1 square-pad/VPP end is resolved from IMG_2437 and corroborated by IMG_2439: 1 VPP, 2 VDD, 3 GND, 4 DAT, 5 CLK. Electrical signal destinations are unchanged. B22-B27 resolve physical D6 upper V to GND and lower W to VPP, each 1 ohm. v0.9.20 B22-B27: upper photo contact V -> GND, lower W -> VPP, each 1 ohm. B29 red V/black W about 0.7V; B30 reverse OL. Supports A=V/GND (pin2), K=W/VPP (pin1). G3 identity and Zener breakdown remain unverified; no 3.9V measurement.

Uncertain: D6 exact identity/class, breakdown voltage and ICSP voltage compatibility remain unknown. The 3.9V value is only a marking candidate, not a measured clamp voltage. J1 geometry remains photo-estimated.

Next: No further D6 routing or polarity checks. Resolve actual breakdown/type only with appropriate part evidence or a controlled isolated test if required; diode mode does not measure breakdown.

I01 - Candidate identities / transistor pin maps

Known: Primary datasheets and marking evidence are recorded; custom symbols use the selected candidate pin tables. D3 measured rail connections fit the BAV99 clamp candidate; its identity is not newly proven.

Uncertain: Short markings do not uniquely prove manufacturers. Candidate choices and some B/E/C or MOSFET source/drain assignments remain qualified. Western alternatives are redesign options, not identities of fitted parts.

Next: Use electrical pad behavior and package/marketing evidence for specific candidates. Do not request sharper photos where the existing marking is already legible.

I02 - LED / motor connector

Known: V127 reaches the photo-selected LED common pair; the two PIC LED drives are mapped. MX512H OUTA/OUTB map to the modeled motor connector. User confirms LED1 contains red and green emitters in a square 1.5 x 1.5 mm body; the schematic now says Red / Green.

Uncertain: Which R30/R31 (PIC22/PIC23) branch lights red versus green, exact die pairing/polarity and manufacturer pin numbering remain unconfirmed. J5 physical A/B lead polarity remains provisional.

Next: Only if channel identification is needed, use diode mode from each resistor-fed LED contact toward its grounded contact and record the emitted colour; do not repeat known board-routing checks. Match J5 wire positions to U5 outputs by continuity.

Values, mechanics and scope

V01 - Unmarked passive values

Known: C5 is 470 uF / 16 V; 38 fitted resistors have nominal marked values. R8 full-body dimensions support 0603 and C11 dimensions support 1206.

Uncertain: Unmeasured fitted ceramic values and L1/L2 type/value remain listed by the component audit. Voltage rating, dielectric, resistor tolerance and power rating are separate evidence gaps.

Next: After one tuning cell topology is verified, prioritize its effective capacitance, RF shaping and receiver filters before bypass values. C11/C12 presently share both modeled nodes: an in-circuit reading is their effective network value, not individual values. Record frequency, mode, amplitude and in-circuit/isolated condition. Separate individual values only when needed for the BOM; do not assign an invented split or blanket 100nF.

M01 - Package geometry / PCB reproduction

Known: 148/149 entries have candidate stock footprints; only LED1 remains blank. C5 is measured 6.33 mm diameter x 16 mm high and uses the nominal 6.3 mm radial footprint. S1 measures 6 x 6 x 4 mm plus 2 mm actuator (6 mm overall). LED1 is red/green, square 1.5 x 1.5 mm. R8/C11 bodies are measured; J1 numbering is corrected.

Uncertain: C5 lead pitch/drill fit remain inferred (stock 2.5 mm/0.8 mm); its stock 7 mm housing differs from the measured 16 mm. LED1 land pattern/numbering remain unresolved. S1 exact pad fit is unverified and stock height is 4.3 mm versus measured 6 mm. Other pitches, drills and board geometry remain partly photo-based.

Next: For PCB reproduction, verify C5 lead spacing and S1 pad fit; use measured C5 16 mm and S1 6 mm heights for clearance. Obtain a matching LED1 vendor footprint/pad drawing. No repeat body measurement is needed.

D01 - Empty options

Known: Two open DNP pads remain: C49.1 and C49.2. R46, U7, R47/C42 and all J6/Q9/R35/R36 pads now have local routes. v0.9.21 annotated photo: J6 left/model1 joins Q9 single/native3 and R36 right/model2. J6 right/model2 joins R36 left/model1. PIC26/RB5-TP10-R35 upper; R35 lower-Q9 paired-left; paired-right via V021 is GND. All J6/Q9/R35/R36 options remain DNP. R36 left/J6 right onward rail is not established by this annotation. U6A is fully mapped: single pad after R37, upper-right J3.1/PIR supply, lower-right GND; DNP retained. v0.9.24 confirms C49 is physically present and empty on the rear in IMG_2442.jpg beside antenna terminations. Options are now drawn with their corresponding functional blocks.

Uncertain: C49 has two untraced pads. J6.2/R36.1 onward identity and intended option values remain unknown. Q9 NPN type and U6A regulator type are circuit-role inferences for empty footprints, not identified fitted devices. U6A now uses a stock regulator role template; U6.3/U106.2 share H_PIR_VDD, so native ERC reports a retained power-output conflict. U6A remains DNP; simultaneous population is not validated.

Next: Keep C49 in the antenna block and resolve its two rear-pad routes from photos first. Keep the Q9/J6 option with PIC outputs and U6A with the PIR regulator. Ask only for hidden continuations that photo review cannot establish.

F01 - Function / internal layers / daughterboard

Known: The one-sheet schematic has a native/model connectivity check and retained photo/via evidence; see the current validation counts.

Uncertain: No powered functional validation, actual layer stack or routed KiCad PCB has been created. The PIR daughterboard internal circuit is not reconstructed. MCU firmware behavior and operating timing are not established.

Next: Finish priority electrical gaps first. Then test defined functions/voltages and decide whether physical PCB reproduction and daughterboard reverse engineering are needed.

Every unrecovered ceramic value

C1, C2, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48

C5 is already 470 uF / 16 V. In-circuit LCR readings may include parallel components and semiconductor paths; record frequency, mode and whether a lead was isolated. Geometry cannot establish capacitance, dielectric or voltage rating.

Closed issues retained as history

E02 - User identifies D3.L=GND and D3.R=VDD through V082. Move only D3.R off the hypothetical receiver supply. D3.S-to-R42 measures 1.2 ohm (resistor end not stated; photo selects pad2). Signal function remains with E06 and candidate identity with I01.

PIC connections

Physical package pins. All have modeled local nets; the evidence column distinguishes confirmed endpoints from photo-derived or incomplete branches. Use the separate numbered photo guide for physical locations.

Pin	Function	Local destination / evidence
1	RE3/MCLR	J1.1/VPP plus D6 cathode; D6 W-VPP 1 ohm, forward 0.7V from GND
2	RA0	C25 signal pad / V031; onward destination unknown
3	RA1	C26 signal pad (user annotation); R28 + R29 + TP7 (earlier meter/trace)
4	RA2	VREF TP via V035; user confirms via-to-TP; C39.1 local photo route
5	RA3	V036-V108 to R14; user continuity report
6	RA4	V037-V102 to R15; user continuity report
7	RA5	V039-V097 to R16; user continuity report
8	VSS	GND / V041 photo anchor
9	RA7/OSC1	Y1 oscillator; photo-derived
10	RA6/OSC2	Y1 oscillator; photo-derived
11	RC0	V047-V091 to R11; user continuity report
12	RC1	R18.1 via V053-V070; user-confirmed; VREF link explicitly rejected
13	RC2	TP3 via V049; user-confirmed; existing R10 branch photo-derived
14	RC3	TP11; visible route; onward destination unknown
15	RC4	V022-V016 to U5 pin2 / INA; user-confirmed
16	RC5	V023-V019 to U5 pin3 / INB; user-confirmed
17	RC6	R13; user meter / visible trace
18	RC7	TP17; visible route; old R13 guess withdrawn
19	VSS	GND / C32 / V045 photo anchor
20	VDD	V042 to board VDD TP user-confirmed; C32.1 local photo route
21	RB0	TP16/Q2.S; user correction. Old VREF join rejected; TP16-VREF 800 kohm
22	RB1	TP2 / R31; user meter / visible trace
23	RB2	TP1 / R30; user meter / visible trace
24	RB3	R4; photo-derived; sensing role remains inferred
25	RB4	TP4 / V017 / R42.1 / R44.1; user continuity; C41 photo-derived
26	RB5	TP10 / R35; user meter / visible trace
27	RB6/CLK	ICSP CLK / D4 / R32; photo-derived
28	RB7/DAT	ICSP DAT / D5 / R40; photo-derived

Every remaining open pad

Populated entries: .

Empty options: C49.1, C49.2.

TP9 was withdrawn as an unsupported inherited inventory entry; rear C49 is photographed and retained. U6 pin4 is unused by user/photo evidence; pin5 externally joins pin2/VIN (1 ohm), although internally NC in the candidate datasheet. Photo pad S means single pad, not necessarily source. The completion checklist includes closure criteria and the full component audit. Sources: reconstruction.json, remaining_work.json, via_audit.json, native validation and preserved measurement records.